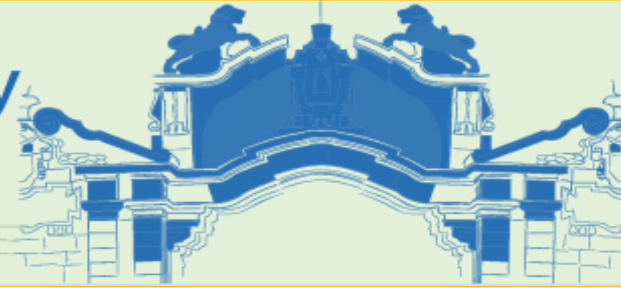




SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !

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Addis Ababa Institute of Technology

School of Mechanical and Industrial Engineering

Engineering Materials I

(MEng 2122)

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July, 2022

Chapter - 6

Mechanical Properties of Metals

Mechanical Properties of Metals

Content Outline

- Introduction
- Stress and strain
- Principal loading conditions
- Methods of testing
- Types of mechanical testing
- Elastic deformation
- Stress and strain behavior
- Plastic deformation
- True stress and strain

Mechanical Properties of Metals

Introduction

- Many materials are subjected to forces or loads when in service.
- The mechanical behavior of a material reflects its response or deformation in relation to an *applied load* or *force*.
- The key mechanical design properties are *stiffness*, *strength*, *hardness*, *ductility*, and *toughness*.
- The mechanical properties of materials are ascertained by performing carefully designed laboratory experiments that replicate as nearly as possible the service conditions.
- Factors to be considered include; *the nature of the applied load* and *its duration*, as well as *the environmental conditions*.

Introduction cont.

- **Application time** may be only a fraction of a second (s), or it may extend over a period of many years.
- **Service temperature** can be an important factor.
- **Mechanical properties** are of concern to a variety of parties that have *differing interests*.
- The most known active and standard materials testing organization is the *American Society for Testing and Materials (ASTM)*.
- The role of **structural engineers** is to determine *stresses* and *stress distributions* within members that are subjected to *well-defined loads*.

Introduction cont.

- **Materials** and **metallurgical engineers**, however, are concerned with *producing* and *fabricating* materials to meet service requirements as predicted by *stress analyses*.
- **Metals** are frequently chosen for *structural applications* because they have desirable combinations of *mechanical characteristics*.
- The *mechanical behavior of metals* is highly considered in this chapter, because *polymers* and *ceramics* are treated separately because they are *to a large degree* and *mechanically different* from **metals**.

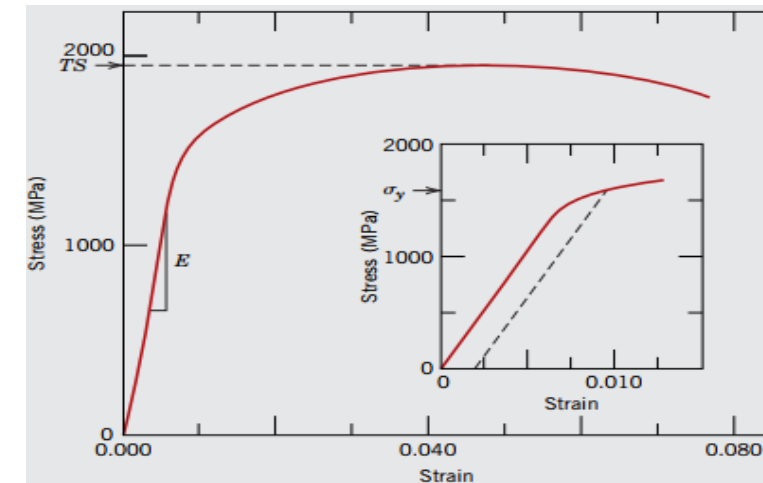


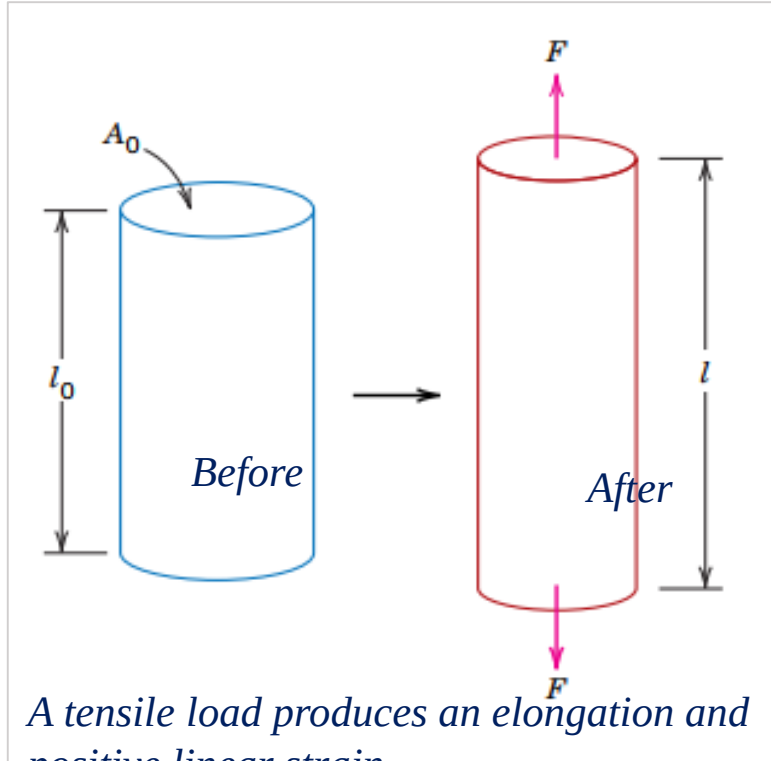
Figure: mechanical testing machine and sample stress-strain diagram.

Concept of Stress and Strain

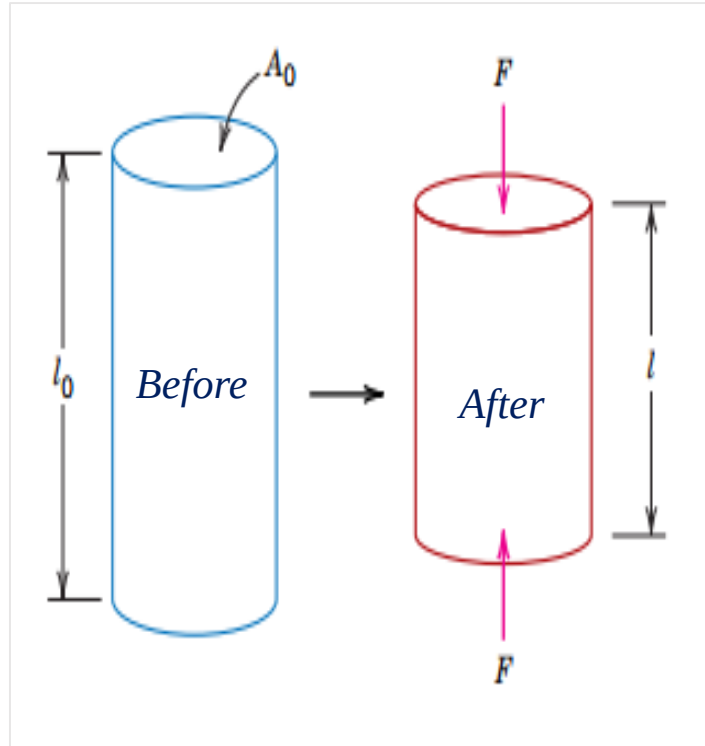
- If a **load** is static or changes relatively slowly with time and is applied uniformly over a cross section or surface of a member, the *mechanical behavior* may be ascertained by a simple *stress–strain test*.
- These are most commonly conducted for **metals** at *room temperature*.
- There are **three principal** ways in which a load may be applied:
 - 1) tension,
 - 2) compression, and
 - 3) shear.
- In engineering practice many loads are *torsional* rather than *pure shear*.

Concept of Stress and Strain - types of loads

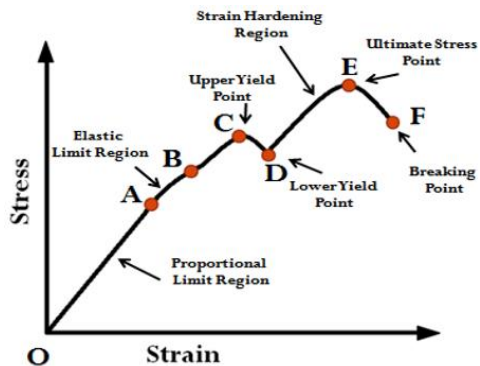
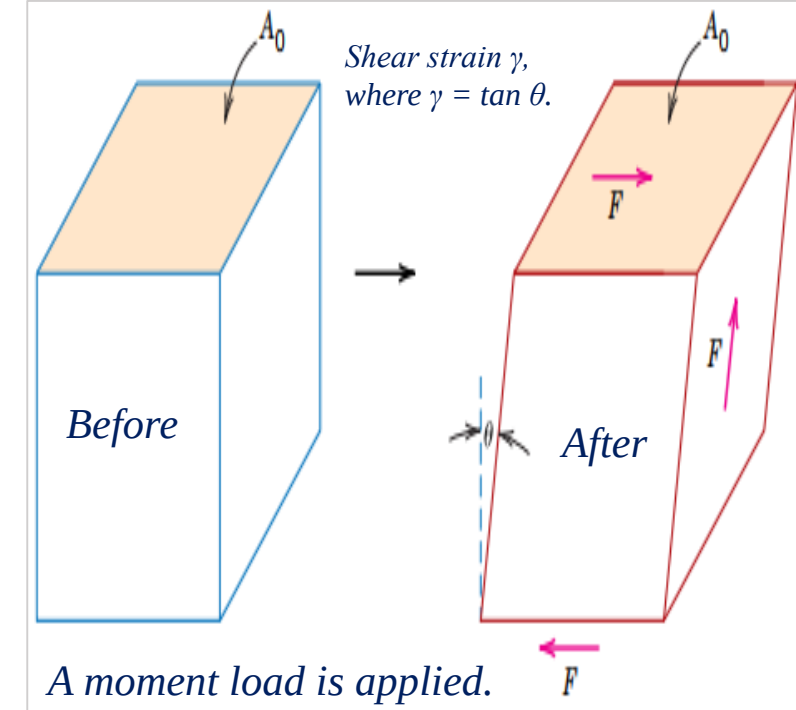
Tension



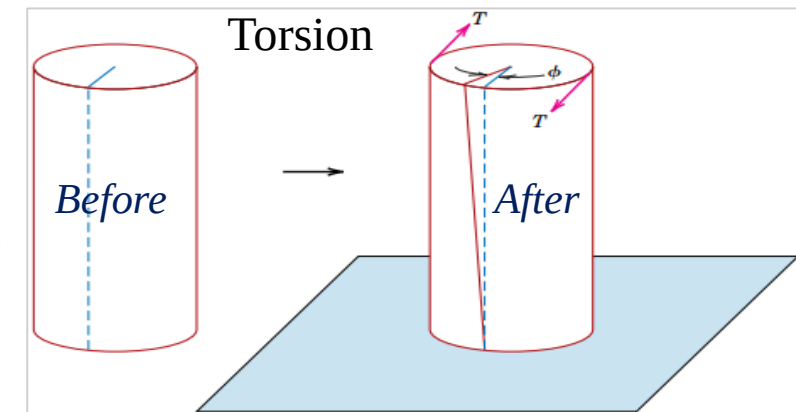
Compression



Shear



Torsion



Stress and Strain

Methods of Materials Testing

- There are **two basic methods** for materials testing;
 - 1) Destructive (mechanical) testing
 - 2) Non-destructive testing
 - **Destructive testing** (DT) is a material testing method as a form of object analysis that involves applying a test to *break down* a particular material to determine its mechanical properties.
 - Some of the common destructive testing are;
 - *tensile testing,*
 - *toughness testing,*
 - *impact testing,*
 - *shear testing, and*
 - *torsional testing,*
 - *fatigue testing.*
- *Properties to be studied; strength, elasticity, plasticity, hardness, toughness, brittleness, stiffness, impact and ductility.*

Stress and Strain

Methods of Materials Testing

- **Non-destructive testing** (NDT) is the process of *testing, inspecting, or evaluating* materials, components or assemblies for properties, discontinuities, or differences in characteristics *without destroying the serviceability* of the part or material.
- The most common NDT are;
 - *visual testing,*
 - *ultrasonic testing,*
 - *radiographic testing,*
 - *eddy current testing,*
 - *magnetic particle testing,*
 - *laser testing,*
 - *liquid penetrant testing, and*
 - *leak testing.*

Note: In this course destructive or mechanical testing is considered to study the mechanical properties of metals.

Stress and Strain

Tensile Tests

- One of the most common *mechanical stress–strain* tests is performed in *tension*.
- The *tensile test* can be used to ascertain several *mechanical properties* of materials that are important in *design*.
- A *specimen* is deformed, usually to fracture, with a gradually *increasing tensile load* that is applied uniaxial along the long axis of a specimen.
- A *standard tensile specimen* is circular cross section, but *rectangular specimens* are also used.

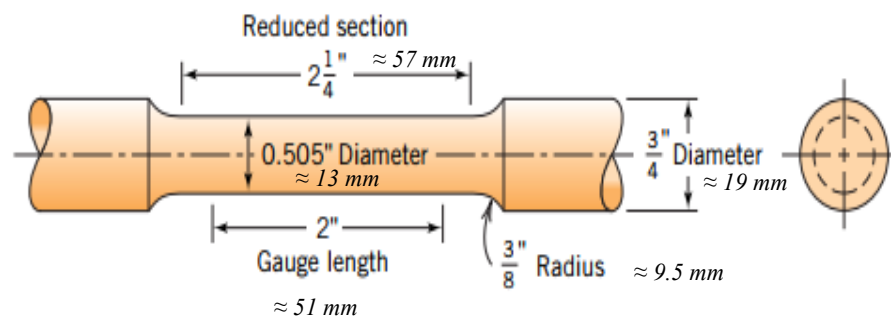


Figure 6.2: A standard tensile specimen with *circular* cross section.

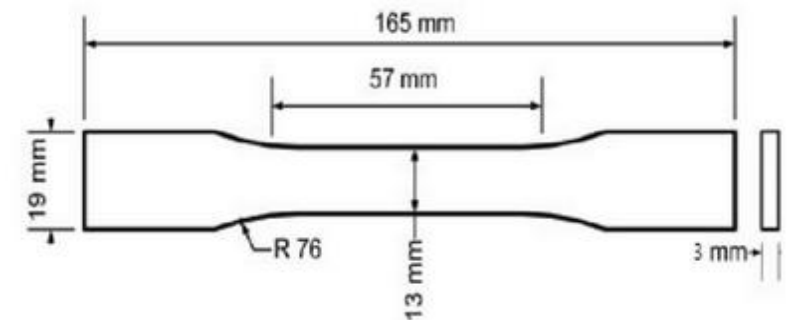


Figure 6.3: A standard tensile specimen with *rectangular* cross section.

Stress and Strain

Tensile Tests

- The output of such a *tensile test* is recorded as *load* or *force* versus *elongation*.
- The *load* vs *deformation* characteristics are depends on the *specimen size*.
- To minimize these *geometrical factors*, *load* and *elongation* are normalized to the respective parameters of *engineering stress* and *engineering strain*.
- **Engineering stress** (σ) is defined by the relationship:

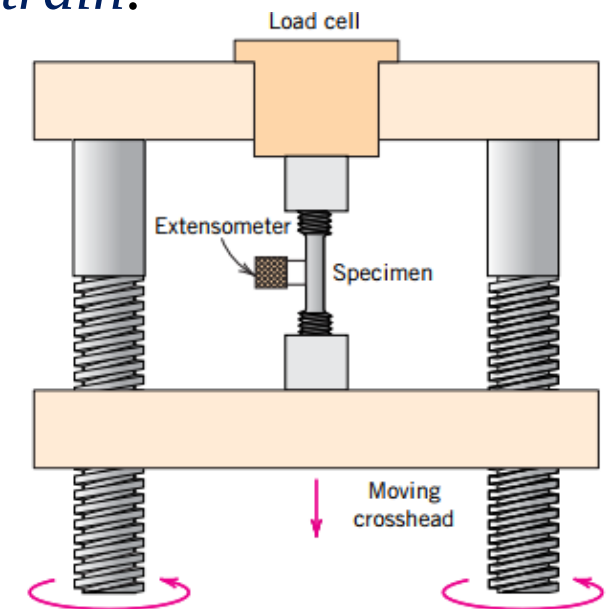
$$\sigma = \frac{F}{A_0}$$

F = is the applied load.
 A_0 = is the original cross-sectional area.

- **Engineering strain** (ϵ) is defined according to:

$$\epsilon = \frac{l_i - l_0}{l_0} = \frac{\Delta l}{l_0}$$

l_0 = is the original length
 l_i = is the instantaneous length and
 Δl = is the change in length.



Tensile test refer to : <https://www.youtube.com/watch?v=uWgnBNOy-rA>
<https://www.youtube.com/watch?v=2jElnzjx-LU>

Stress and Strain

Compressive Tests

- A *compressive test* is conducted in a manner similar to the *tensile test*, except that the force is compressive and the specimen contracts along the direction of the stress.
- A *compressive force* is taken to be negative, which yields a negative stress. This is because l_0 is greater than l_i .
- *Tensile tests* are more common than *compressive tests* in different structural applications, because very little additional information is obtained from *compressive tests*.
- *Compressive tests* are used when a material's behavior under large and permanent (i.e., plastic) strains is desired.

Stress and Strain

Shear and Torsional Tests

- The **shear stress** (τ) is computed using a pure shear force (F) according to:

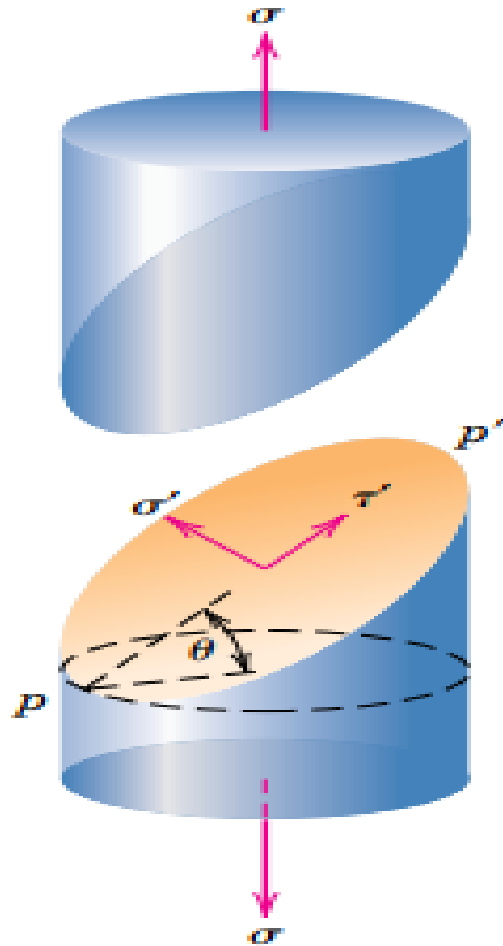
$$\tau = \frac{F}{A_0} \quad A_0 = \text{is the cross section.}$$

- The **shear strain**, γ is defined as the tangent of the strain angle θ .
- **Torsion** is a variation of pure shear in which a structural member is twisted in the manner of torsional forces produce a rotational motion about the longitudinal axis of one end of the member relative to the other end.
- **Torsional** tests are performed on *cylindrical solid shafts* or *tubes*.
- **Shear stress**, τ also can be expressed as a function of the applied torque T .

$$\tau = \frac{T r}{J}$$

Stress and Strain

Shear and Torsional Tests



According to stress system in mechanics-of-materials principles;

$$\sigma' = \sigma \cos^2 \theta = \sigma \left(\frac{1 + \cos 2\theta}{2} \right)$$

$$\tau' = \sigma \sin \theta \cos \theta = \sigma \left(\frac{\sin 2\theta}{2} \right)$$

Or,
Based on the vector decomposition;

$$\sigma' = \sigma \cos \theta$$

$$\tau' = \sigma \sin \theta$$

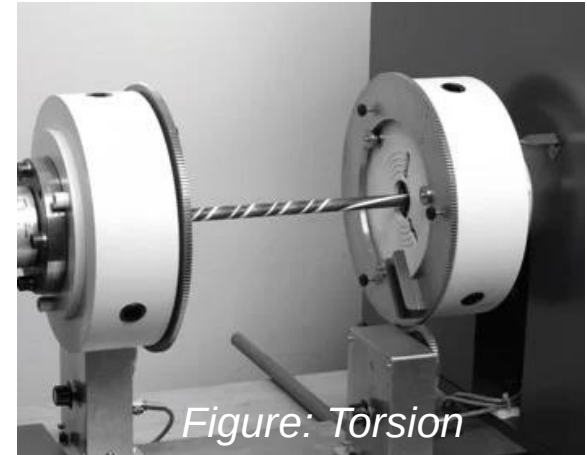
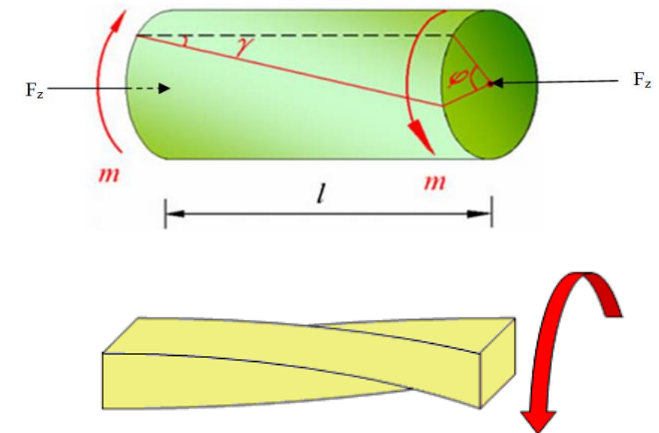


Figure: Torsion



Elastic Deformation

Stress-Strain Behavior

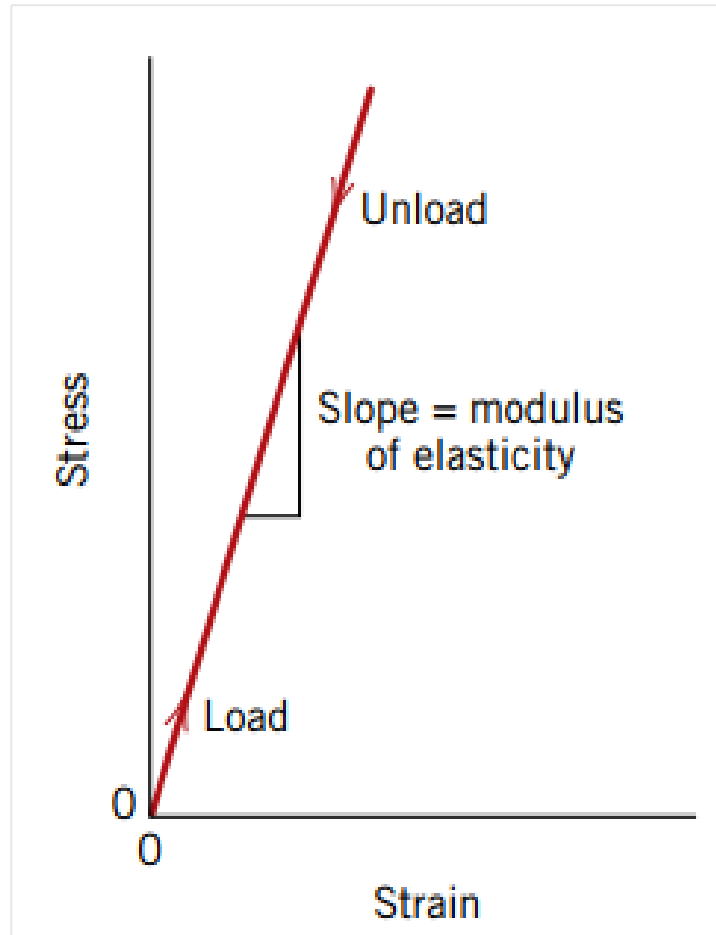
- Deformation in which stress and strain are proportional is called *elastic deformation*.
- *Hooke's law* - relationship between engineering stress and engineering strain for elastic deformation, tension and compression:

$$\sigma = E\epsilon \quad \rightarrow \text{Hooke's law}$$

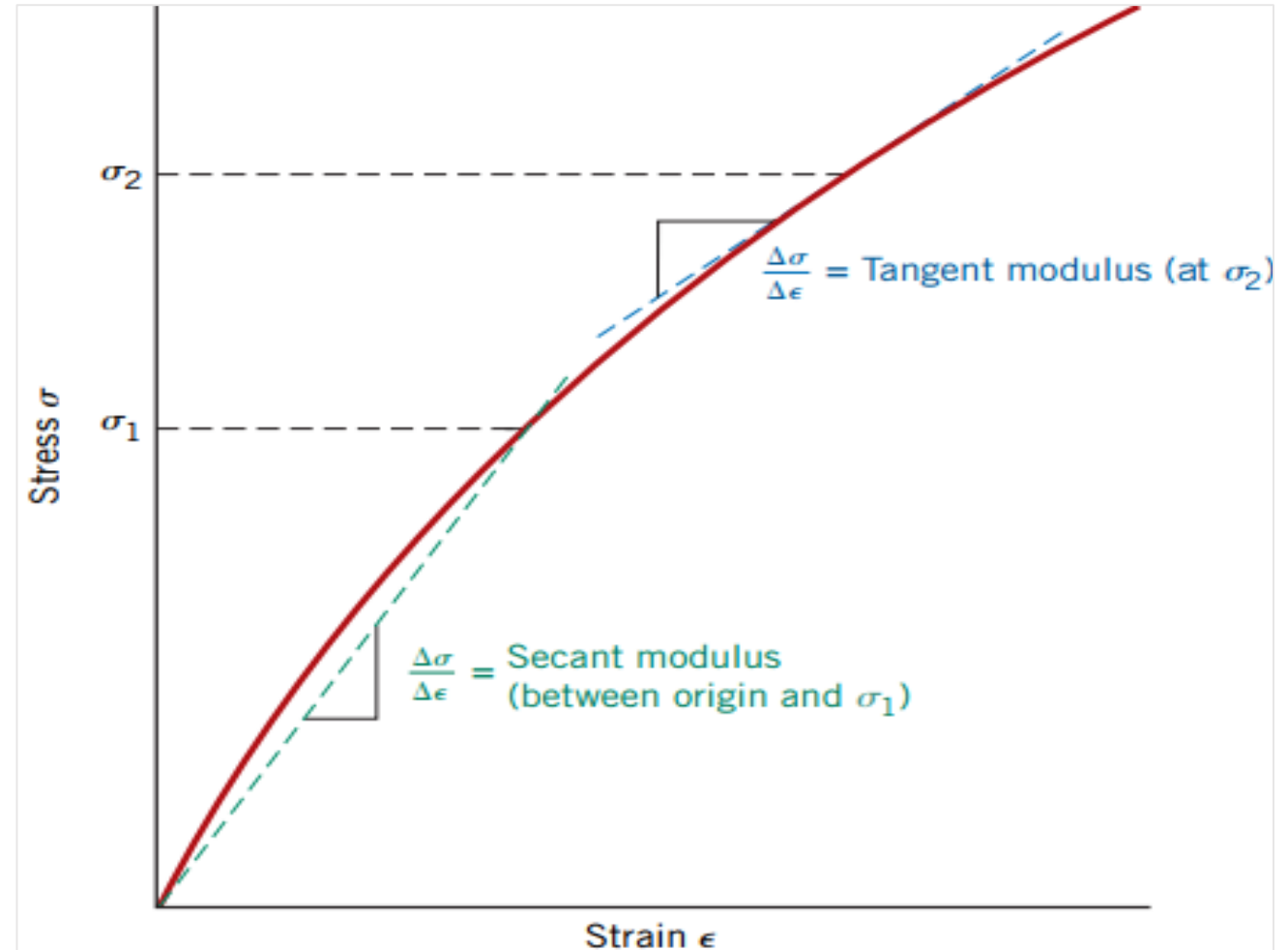
- The *modulus of elasticity* (E) is the constant of proportionality.
- It is also called *Young's modulus* and its SI unit is GPa or psi, $45 \text{ GPa} = 6.5 * 10^6 \text{ psi}$.
- The slope of this linear segment corresponds to the *modulus of elasticity* (E).
- The *greater* the modulus, the *stiffer* the material.
- The *modulus* is an important design parameter for computing *elastic deflections*.

Elastic Deformation

Stress-Strain Behavior



Stress–Strain diagram → for linear elastic deformation.



Stress–Strain diagram → for nonlinear elastic deformation.

Elastic Deformation

Stress-Strain Behavior

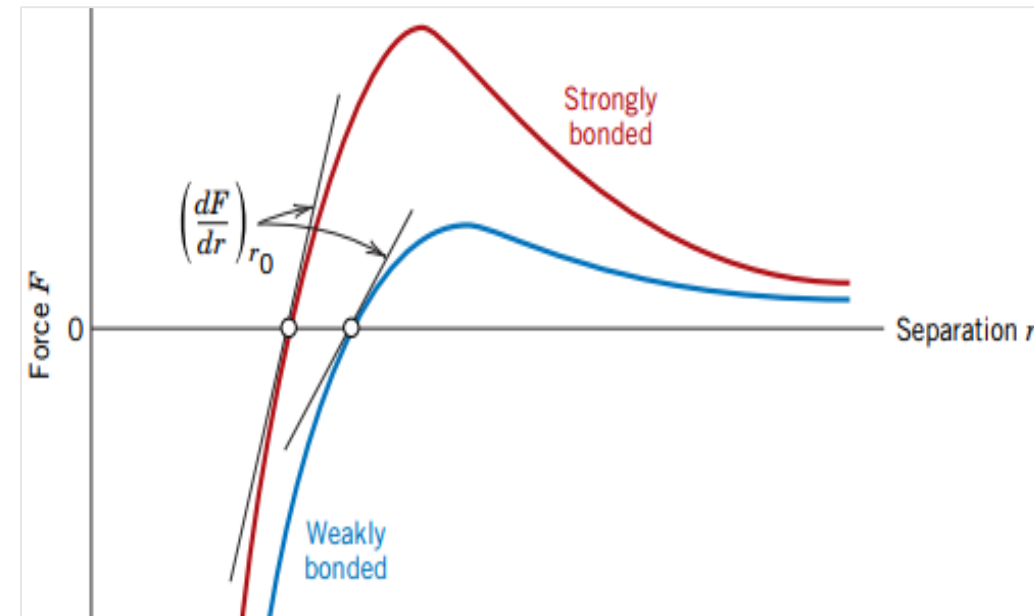
- The **modulus** is proportional to the slope of the interatomic force separation curve at the equilibrium spacing:

$$E \propto \left(\frac{dF}{dr} \right)_{r_0}$$

- *Macroscopic elastic strain* is manifested as small changes in the interatomic spacing and the stretching of interatomic bonds.

- The **tangent modulus** is taken as the slope of the stress–strain curve.

- Values of the **modulus of elasticity** for **ceramic** materials and **metals** are higher than **polymers**.



Elastic Deformation

Stress-Strain Behavior

- The **modulus of elasticity** (E) may vary from material to material as a function of temperature. This can be summarized in the figure below.

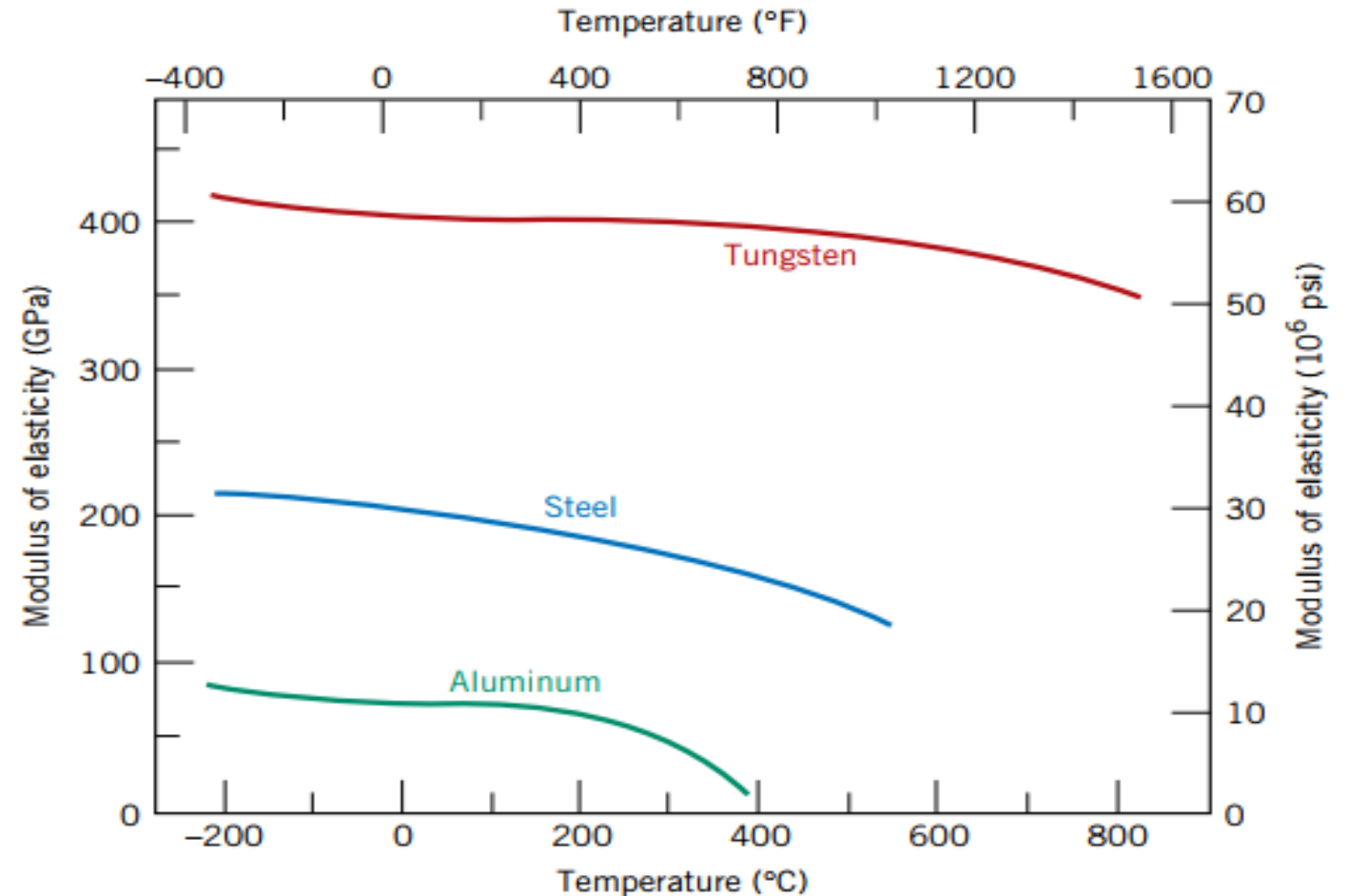


Figure: Plot of modulus of elasticity versus temperature for tungsten, steel, and aluminum.

Elastic Deformation

Stress-Strain Behavior

<i>Metal Alloy</i>	<i>Modulus of Elasticity</i>		<i>Shear Modulus</i>		<i>Poisson's Ratio</i>
	<i>GPa</i>	<i>10⁶ psi</i>	<i>GPa</i>	<i>10⁶ psi</i>	
Aluminum	69	10	25	3.6	0.33
Brass	97	14	37	5.4	0.34
Copper	110	16	46	6.7	0.34
Magnesium	45	6.5	17	2.5	0.29
Nickel	207	30	76	11.0	0.31
Steel	207	30	83	12.0	0.30
Titanium	107	15.5	45	6.5	0.34
Tungsten	407	59	160	23.2	0.28

- Elastic deformation is *nonpermanent*, which means that when the applied load is released, the piece returns to its original shape.

Elastic Deformation

Stress-Strain Behavior

- The **stress–strain** characteristics at low stress levels are virtually the same for both tensile and compressive situations.
- The relationship between **shear stress** and **shear strain** is:

$$\tau = G\gamma$$

G = is the shear modulus, the slope of the linear elastic region of the shear stress–strain curve.

- **Anelasticity** is a time-dependent elastic behavior. That is, elastic deformation will continue after the stress application, and upon load release some finite time is required for complete recovery
- For **metals**, the **anelastic component** is normally **small** and is often neglected.
- For some **polymeric materials**, its magnitude is significant and termed as **viscoelastic behavior**.

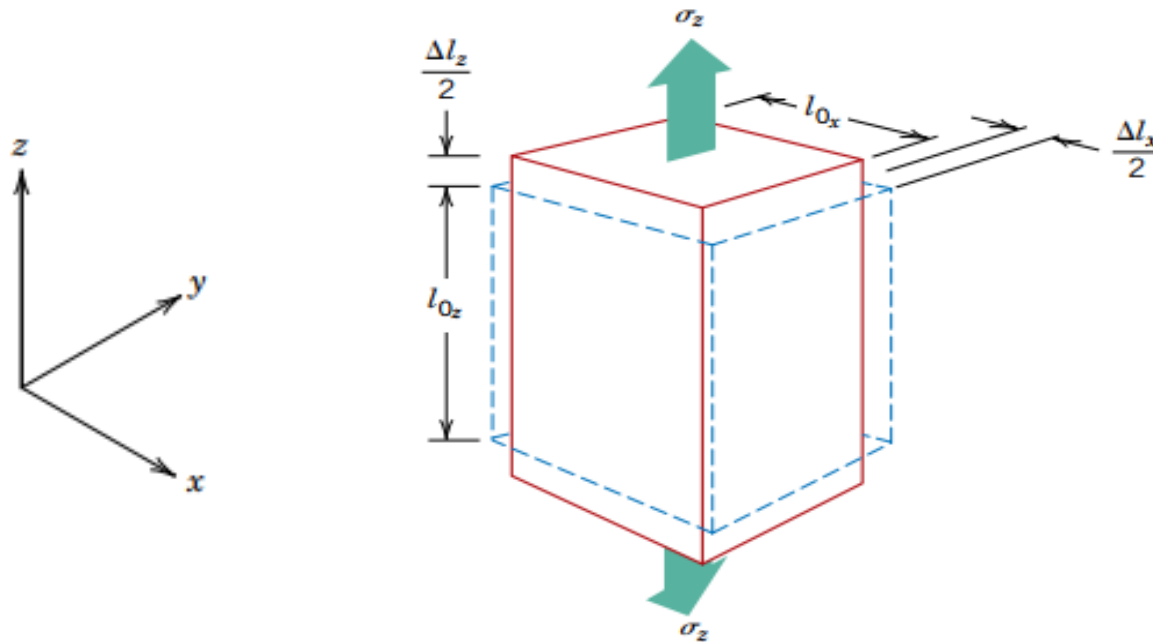
Elastic Deformation

Elastic Properties of Materials

- **Poisson's ratio (ν)**: is defined as the ratio of the *lateral* and *axial* strains.

$$\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\epsilon_y}{\epsilon_z}$$

If the applied stress is uniaxial (only in the z direction) and the material is isotropic, then $\epsilon_x = \epsilon_y$.



$$\frac{\epsilon_z}{2} = \frac{\Delta l_z/2}{l_{0z}}$$
$$-\frac{\epsilon_x}{2} = \frac{\Delta l_x/2}{l_{0x}}$$

Relationship among elastic parameters—modulus of elasticity, shear modulus, and Poisson's ratio:

$$E = 2G(1 + \nu)$$

In most metals, G is about $0.4E$

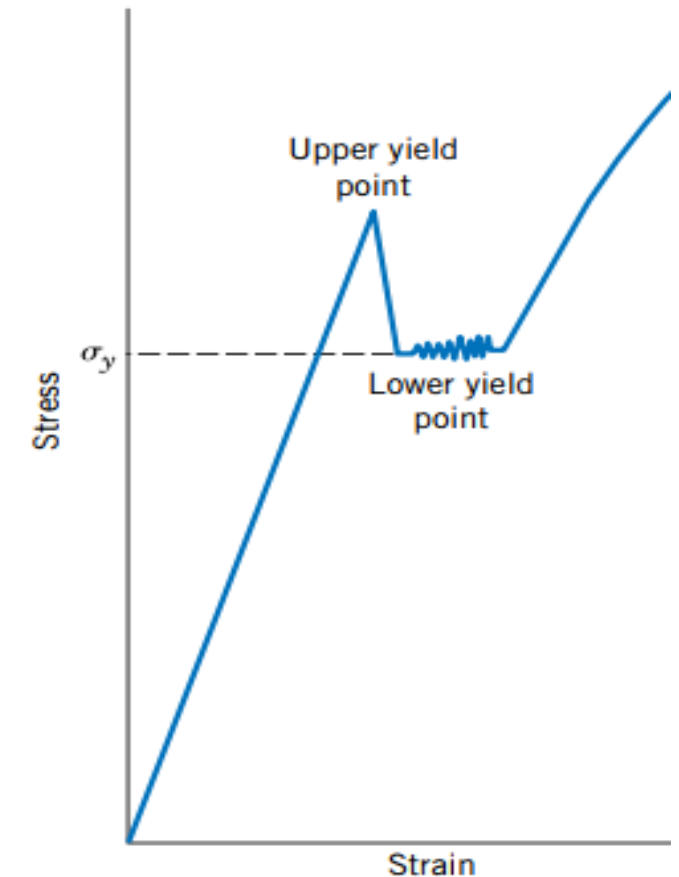
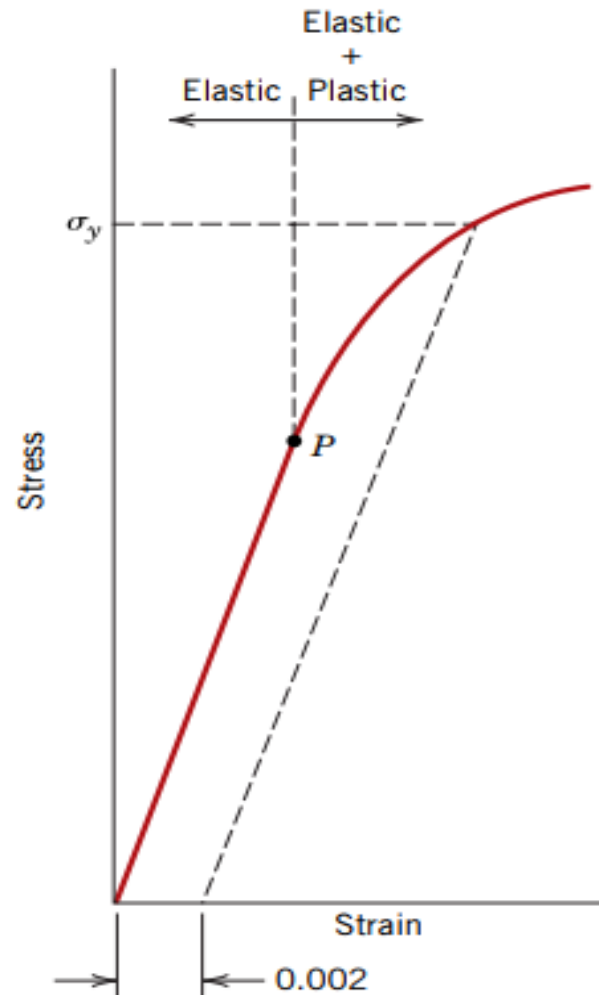
Plastic Deformation

- Plastic (permanent) deformation is a type of deformation at which the stress is no longer proportional to the strain. → *Hooke's law* does not obey.
- The mechanism of this deformation is different for *crystalline* and *amorphous* materials.
- For *crystalline solids*, deformation is accomplished by means of a process called *slip*, which involves the motion of dislocations.
- Plastic deformation in noncrystalline solids (as well as liquids) occurs by a viscous flow mechanism.
- Plastic deformation corresponds to the breaking of bonds with original atom neighbors.

Plastic Deformation

Tensile Properties: yielding and yield strength

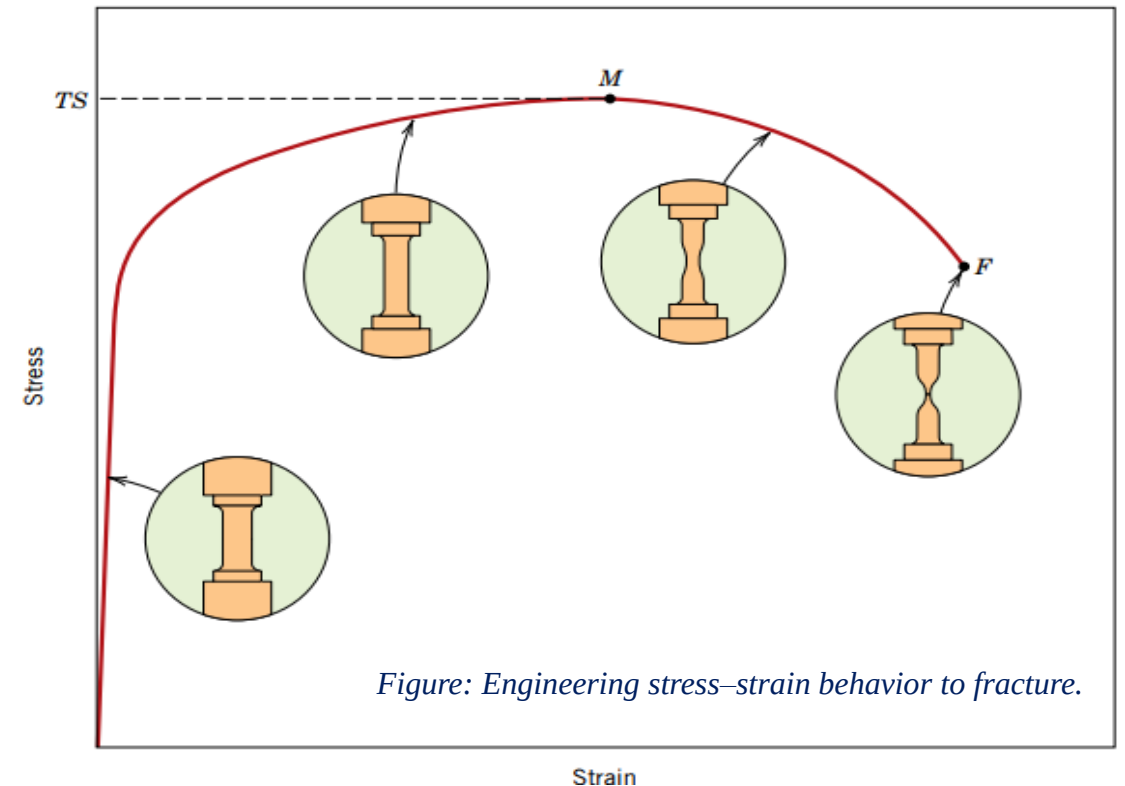
- A structure or component that has plastically deformed—or experienced a permanent change in shape.
- It is important to know the stress level at which plastic deformation begins. This is called **yielding**.
- The magnitude of the **yield strength** for a metal is a measure of its resistance to plastic deformation.



Plastic Deformation

Tensile Properties: yielding and yield strength

- **Yield strengths** may range from 35 MPa for a low-strength aluminum to greater than 1400 MPa for high-strength steels.
- After **yielding**, the stress necessary to continue plastic deformation in metals increases to a maximum.
- The **tensile strength** TS (MPa or psi) is the stress at the maximum on the **engineering stress–strain curve**.



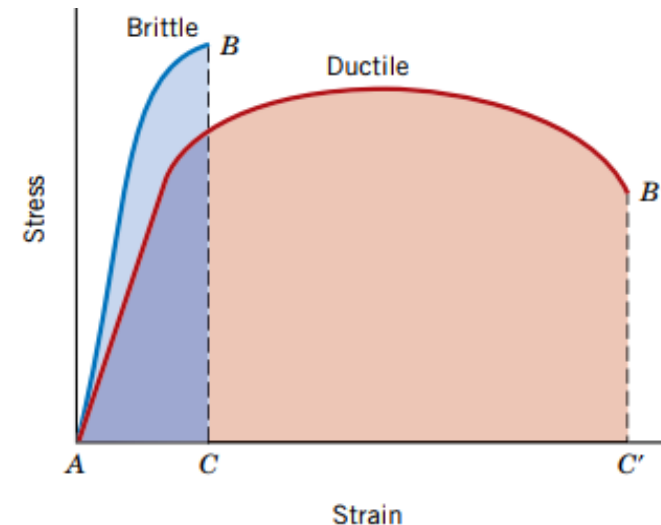
Plastic Deformation

Tensile Properties: yielding and yield strength

- **Ductility** is another important mechanical property.
- It is a measure of the degree of plastic deformation that has been sustained at fracture.
- A metal that experiences very little or no plastic deformation upon fracture is termed *brittle*.
- The *tensile stress–strain* behaviors for both *ductile* and *brittle* metals are very different.
- **Ductility** may be expressed quantitatively as either *percent elongation* or *percent reduction in area*.

$$\%EL = \left(\frac{l_f - l_0}{l_0} \right) \times 100$$

$$\%RA = \left(\frac{A_0 - A_f}{A_0} \right) \times 100$$



Plastic Deformation

Tensile Properties: yielding and yield strength

<i>Metal Alloy</i>	<i>Yield Strength, MPa (ksi)</i>	<i>Tensile Strength, MPa (ksi)</i>	<i>Ductility, %EL [in 50 mm (2 in.)]</i>
Aluminum	35 (5)	90 (13)	40
Copper	69 (10)	200 (29)	45
Brass (70Cu–30Zn)	75 (11)	300 (44)	68
Iron	130 (19)	262 (38)	45
Nickel	138 (20)	480 (70)	40
Steel (1020)	180 (26)	380 (55)	25
Titanium	450 (65)	520 (75)	25
Molybdenum	565 (82)	655 (95)	35

Brittle materials are *approximately* considered to be those having a fracture strain of less than about 5%.

Plastic Deformation

Tensile Properties: yielding and yield strength

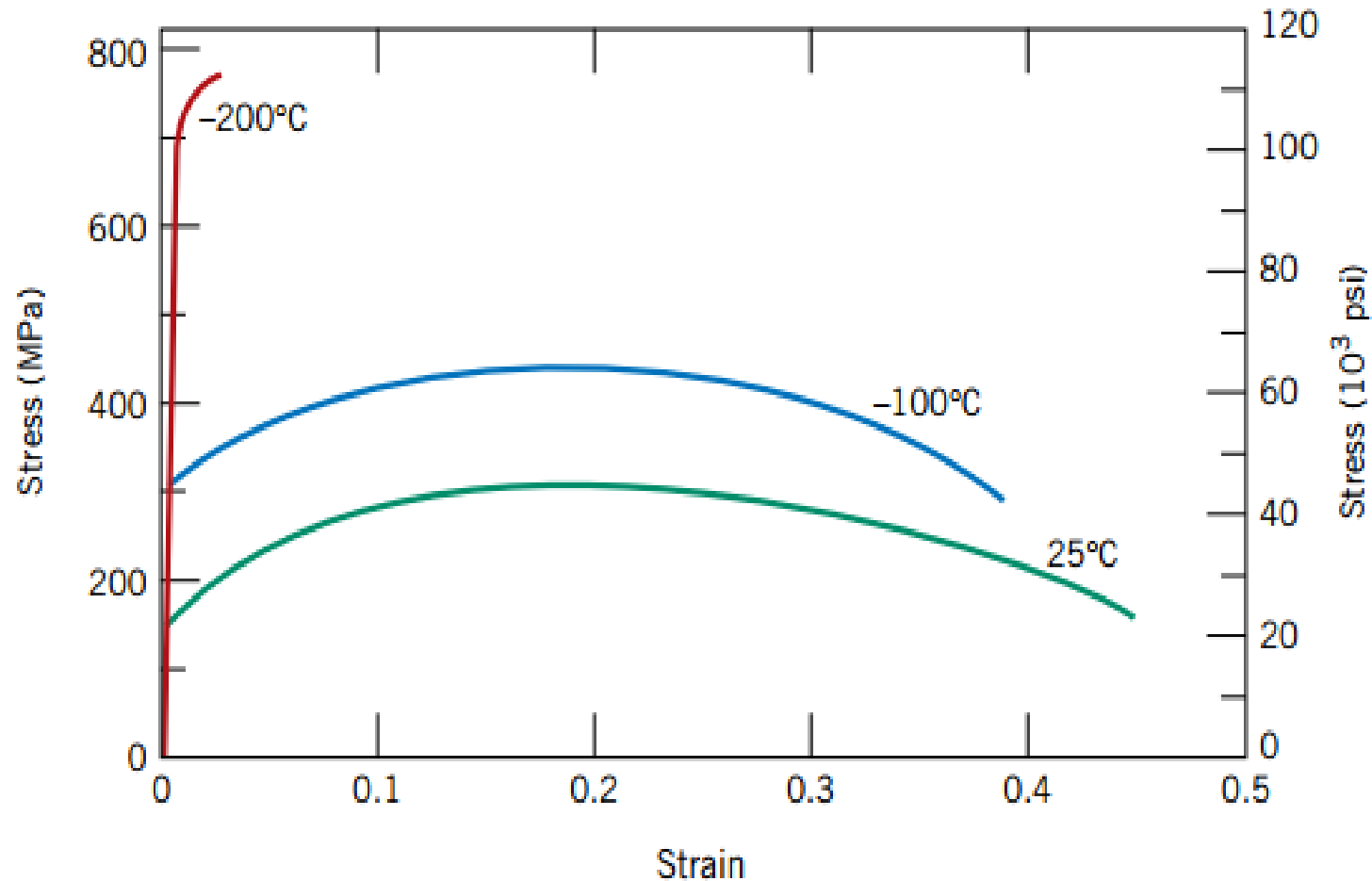
- The **magnitudes** of both yield and tensile strengths decline with increasing temperature.
- **Resilience** is the capacity of a material to absorb energy when it is deformed elastically and then upon unloading, to have this energy recovered.
- The associated property is the **modulus of resilience**, U_r , which is the strain energy per unit volume required to stress a material from an unloaded state up to the point of yielding.
- **Modulus of resilience:**

$$U_r = \int_0^{\epsilon_y} \sigma d\epsilon$$

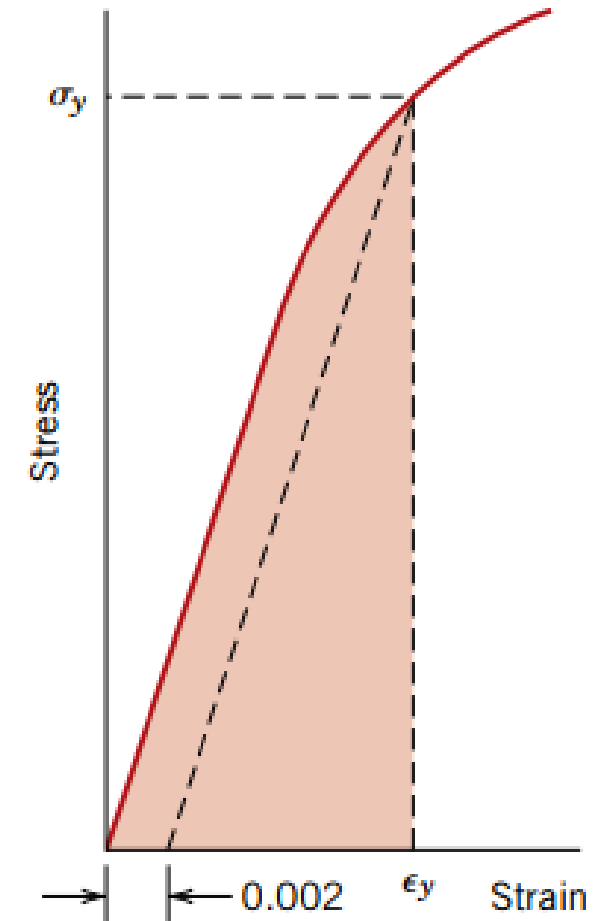
$$U_r = \frac{1}{2} \sigma_y \epsilon_y \rightarrow \text{a linear elastic region.}$$

Plastic Deformation

Tensile Properties: yielding and yield strength



Engineering stress–strain behavior for iron at three temperatures.



Modulus of resilience is determined from the tensile stress–strain behavior of a material.

Plastic Deformation

Tensile Properties: yielding and yield strength

- The SI unit of **resilience**: joules per cubic meter (J/m³, equivalent to Pa).

Modulus of resilience for
linear elastic behavior, and
incorporating Hooke's law:

$$U_r = \frac{1}{2} \sigma_y \epsilon_y = \frac{1}{2} \sigma_y \left(\frac{\sigma_y}{E} \right) = \frac{\sigma_y^2}{2E}$$

- **Toughness** is the ability of a metal to absorb energy during plastic deformation before fracture.
- **Toughness** is a mechanical term that may be used in several contexts.

<i>Material</i>	<i>Yield Strength (MPa)</i>	<i>Tensile Strength (MPa)</i>	<i>Strain at Fracture</i>	<i>Fracture Strength (MPa)</i>	<i>Elastic Modulus (GPa)</i>
A	310	340	0.23	265	210
B	100	120	0.40	105	150
C	415	550	0.15	500	310
D	700	850	0.14	720	210
E	Fractures before yielding			650	350

Plastic Deformation

True Stress and Strain

- **True stress** σ_T is defined as the load F divided by the instantaneous cross-sectional area A_i over which deformation is occurring:

$$\sigma_T = \frac{F}{A_i}$$

- **True strain** ϵ_T , defined by:

$$\epsilon_T = \ln \frac{l_i}{l_0} \quad A_i l_i = A_0 l_0$$

- Conversion of engineering stress to true stress:

$$\sigma_T = \sigma(1 + \epsilon)$$

- Conversion of engineering strain to true strain:

$$\epsilon_T = \ln(1 + \epsilon)$$

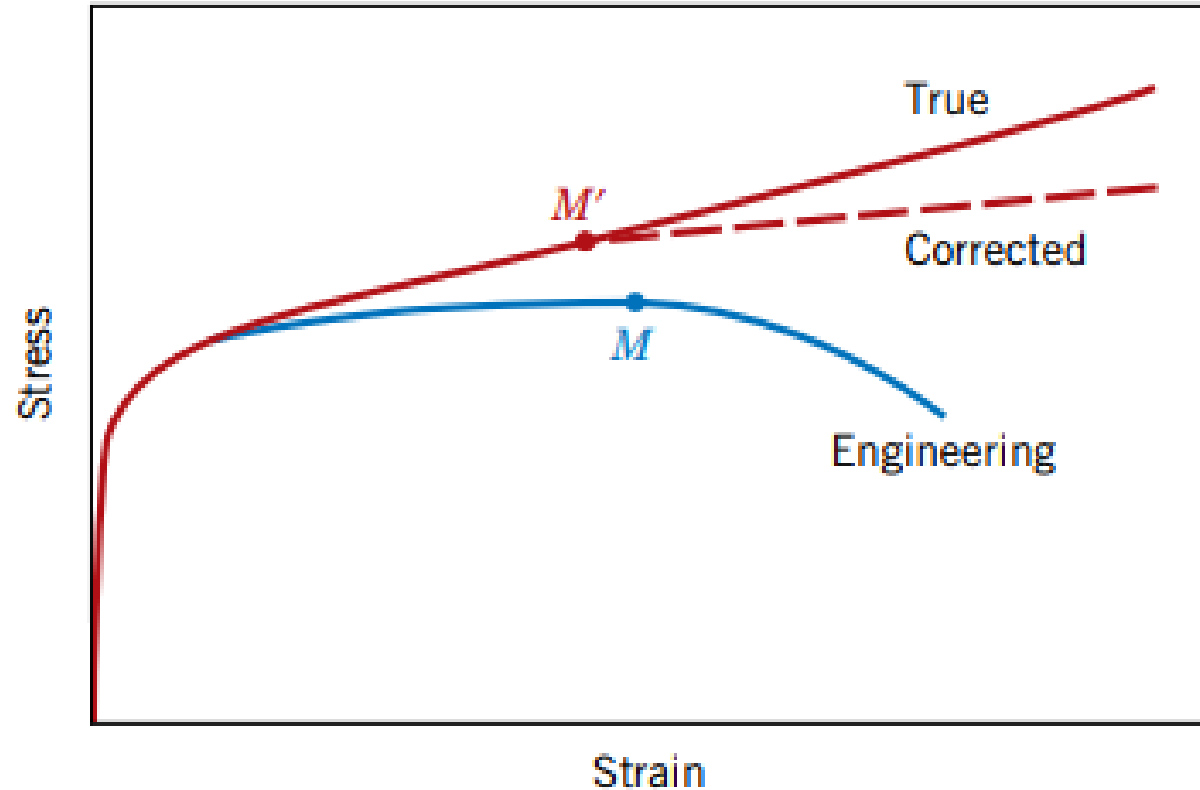
*K & n are constants
vary depend on the
alloy.*

- True stress–true strain relationship in the plastic region of deformation:

$$\sigma_T = K\epsilon_T^n$$

Plastic Deformation

True Stress and Strain



A comparison of typical tensile engineering stress–strain and true stress–strain behaviors. Necking begins at point M on the engineering curve, which corresponds to M' on the true curve. The “corrected” true stress–strain curve takes into account the complex stress state within the neck region.

Figure: A comparison of typical tensile engineering stress–strain and true stress–strain behaviors.

Plastic Deformation

Elastic Recovery after Plastic Deformation

Schematic tensile stress–strain diagram showing the phenomena of elastic strain recovery and strain hardening.

The initial yield strength is designated as σ_{yi} ; σ_{yo} after releasing the load at point D, and then upon reloading

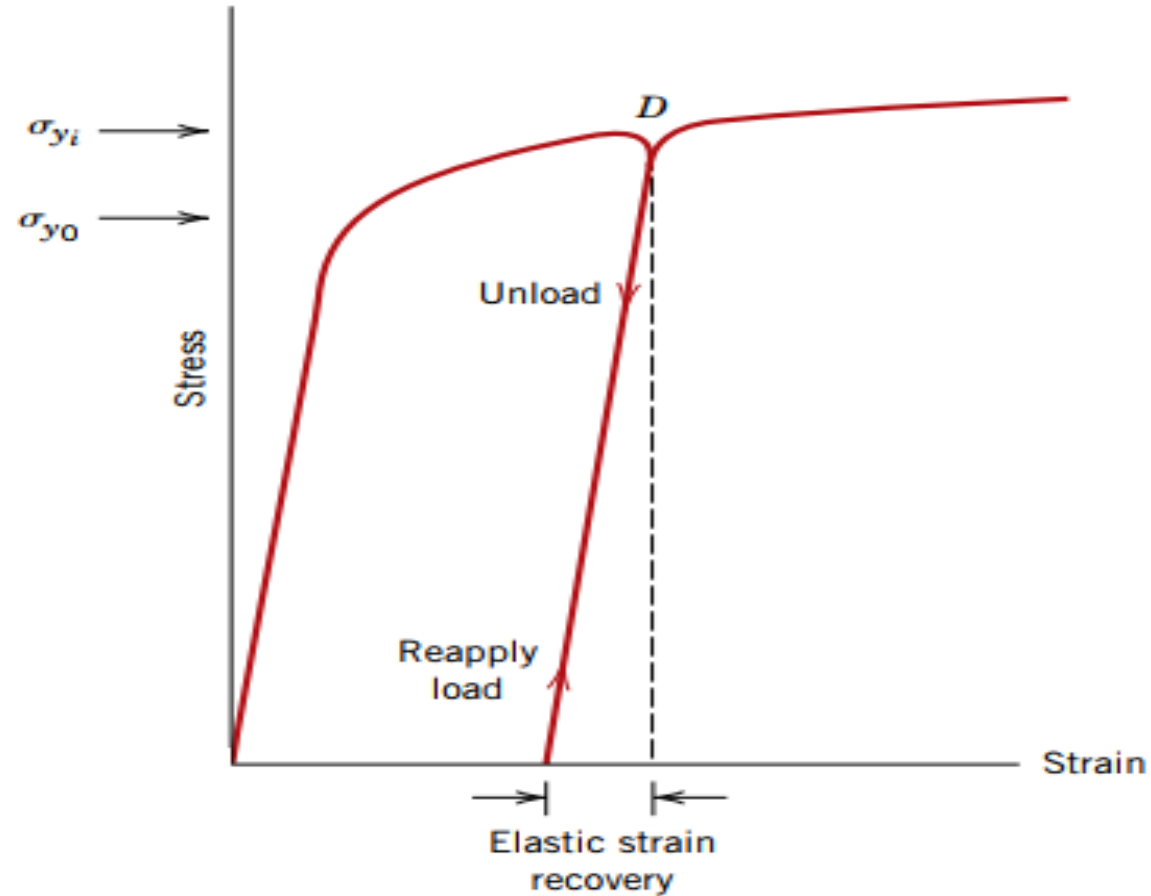


Figure: The phenomena of elastic strain recovery and strain hardening.

Plastic Deformation

Compressive, Shear and Torsional Deformations

- **Metals** may experience plastic deformation under the influence of applied *compressive*, *shear*, and *torsional loads*.
- The resulting **stress–strain** behavior into the plastic region is **similar** to the tensile counterpart.
- However, for compression, there is no maximum because necking does not occur; furthermore, the mode of **fracture** is different from that for tension.

Plastic Deformation

Hardness

- **Hardness** is a measure of a material's resistance to localized plastic deformation.
Example, a small dent or a scratch.
- A qualitative → arbitrary hardness indexing, termed the *Mohs scale*, which ranged from 1 to 10 for diamond.
- Quantitative hardness techniques → the depth or size of the resulting indentation is measured and related to a hardness number;
 - the softer the material,
 - the larger and deeper the indentation, and
 - the lower the hardness index number.

Plastic Deformation

Hardness

- **Hardness tests** are performed more **frequently** than any other mechanical test for several reasons:
 - 1) they are **simple** and **inexpensive**.
 - 2) the test is **nondestructive** — a small indentation is the only deformation.
 - 3) other mechanical properties often may be estimated from hardness data, such as **tensile strength**.

Rockwell Hardness Tests

- The **Rockwell tests** constitute the **most common method** used to measure hardness because they are so simple to perform and require no special skills.

Plastic Deformation

Hardness

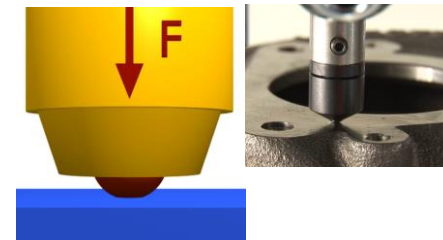
- A **hardness number** is determined by the difference in depth of penetration resulting from the application of an initial minor load followed by a larger major load.
- Utilization of a **minor load** enhances test accuracy.
- On the basis of the magnitude of both **major** and **minor** loads, there are two types of tests: *Rockwell* and *superficial Rockwell*.
- **Rockwell** and **superficial hardnesses**, both hardness number and scale symbol must be indicated.
- The **scale** is designated by the symbol HR.
- Modern **Rockwell hardness** measurements are automated and very simple to use.

Plastic Deformation

Hardness

Brinell Hardness Tests

- In **Brinell tests**, as in Rockwell measurements, a hard, spherical indenter is forced into the surface of the metal to be tested.
- The **diameter** of the hardened steel (or tungsten carbide) indenter is 10.00 mm.
- **Standard loads range** between 500 and 3000 kg in 500-kg increments.
- The **load** is maintained constant for a specified time (between 10 and 30 s). Harder materials require greater applied loads.
- The **Brinell hardness** number (HB), is a function of both the magnitude of the load and the diameter of the resulting indentation.

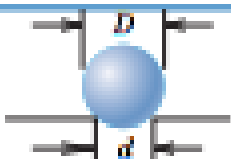
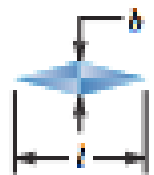
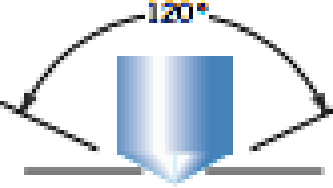
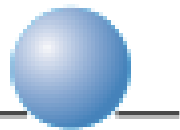


Brinell Hardness Tests

Plastic Deformation

Hardness

Table 6.5 Hardness-Testing Techniques

Test	Indenter	Shape of Indentation		Load	Formula for Hardness Number ^a
		Side View	Top View		
Brinell	10-mm sphere of steel or tungsten carbide			P	$HB = \frac{2P}{\pi D [D - \sqrt{D^2 - d^2}]}$
Vickers microhardness	Diamond pyramid			P	$HV = 1.854P/d_1^2$
Knoop microhardness	Diamond pyramid			P	$HK = 14.2P/b^2$
Rockwell and superficial Rockwell	Diamond cone: $\frac{1}{16}$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$ in. diameter steel spheres	 	 	$\left. \begin{array}{l} 60 \text{ kg} \\ 100 \text{ kg} \\ 150 \text{ kg} \end{array} \right\} \text{Rockwell}$ $\left. \begin{array}{l} 15 \text{ kg} \\ 30 \text{ kg} \\ 45 \text{ kg} \end{array} \right\} \text{Superficial Rockwell}$	

Plastic Deformation

Hardness

Table 6.6a Rockwell Hardness Scales

<i>Scale Symbol</i>	<i>Indenter</i>	<i>Major Load (kg)</i>
A	Diamond	60
B	$\frac{1}{16}$ -in. ball	100
C	Diamond	150
D	Diamond	100
E	$\frac{1}{8}$ -in. ball	100
F	$\frac{1}{16}$ -in. ball	60
G	$\frac{1}{16}$ -in. ball	150
H	$\frac{1}{8}$ -in. ball	60
K	$\frac{1}{8}$ -in. ball	150

Table 6.6b Superficial Rockwell Hardness Scales

<i>Scale Symbol</i>	<i>Indenter</i>	<i>Major Load (kg)</i>
15N	Diamond	15
30N	Diamond	30
45N	Diamond	45
15T	$\frac{1}{16}$ -in. ball	15
30T	$\frac{1}{16}$ -in. ball	30
45T	$\frac{1}{16}$ -in. ball	45
15W	$\frac{1}{8}$ -in. ball	15
30W	$\frac{1}{8}$ -in. ball	30
45W	$\frac{1}{8}$ -in. ball	45

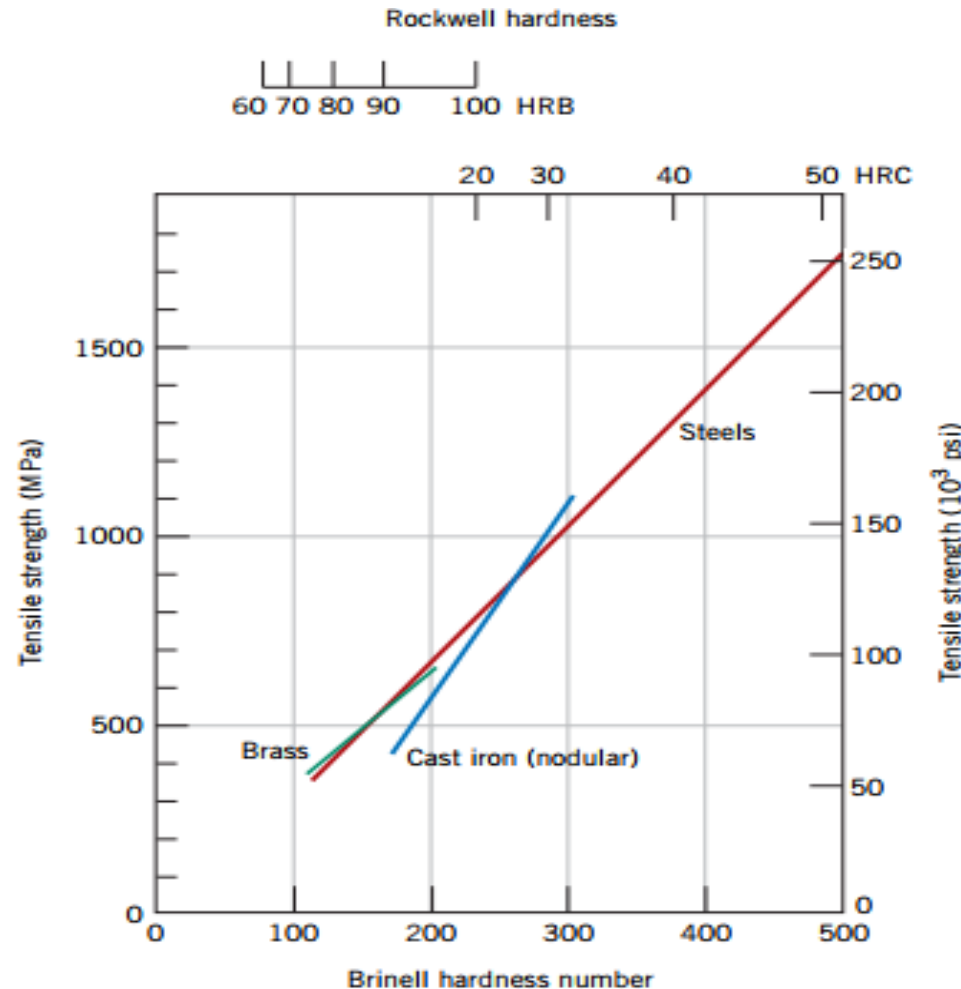
- Semi-automatic techniques for measuring Brinell hardness are available.
- Maximum specimen thickness and indentation position (relative to specimen edges) as well as minimum indentation spacing requirements are the same as for *Rockwell* tests.

Plastic Deformation

Hardness

The difference between Rockwell and Brinell hardness tests:

- *Brinell hardness (HB)* is generally used for the soft material, such as non-ferrous metals, steel before heat treated or after annealing.
- *Rockwell hardness (HRC)* is generally used for high hardness materials, such as the hardness after heat treatment and so on.



Correlation between Hardness and Tensile Strength

For steel alloys, conversion of Brinell hardness to tensile strength:

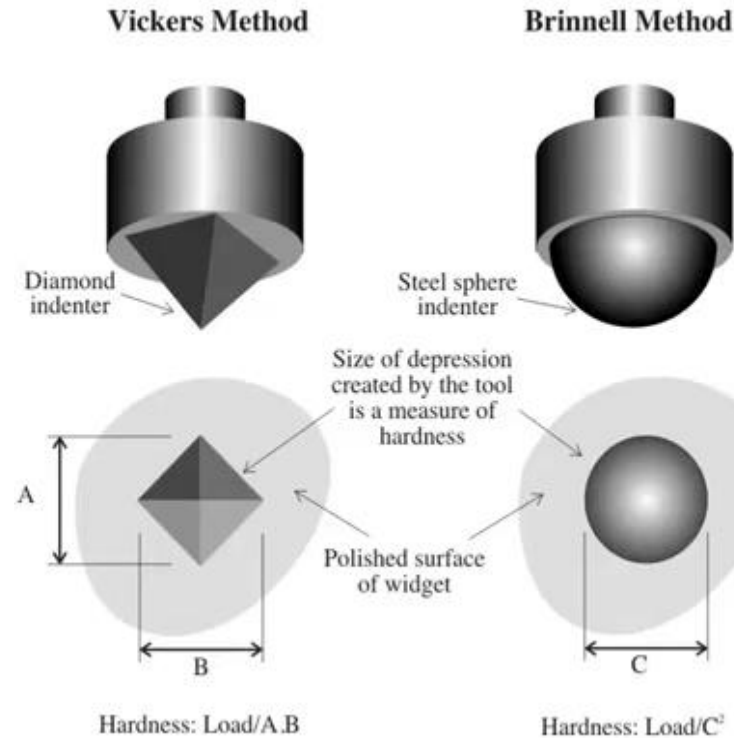
$$TS(\text{MPa}) = 3.45 \times HB$$

$$TS(\text{psi}) = 500 \times HB$$

Plastic Deformation

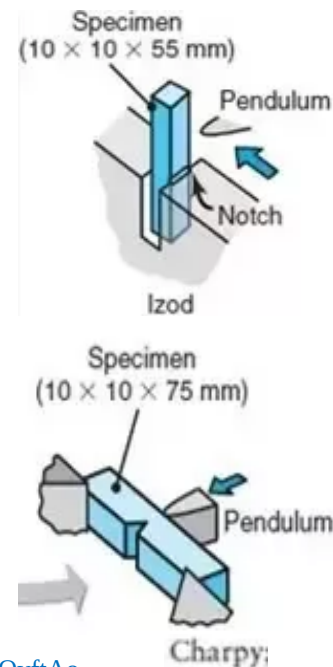
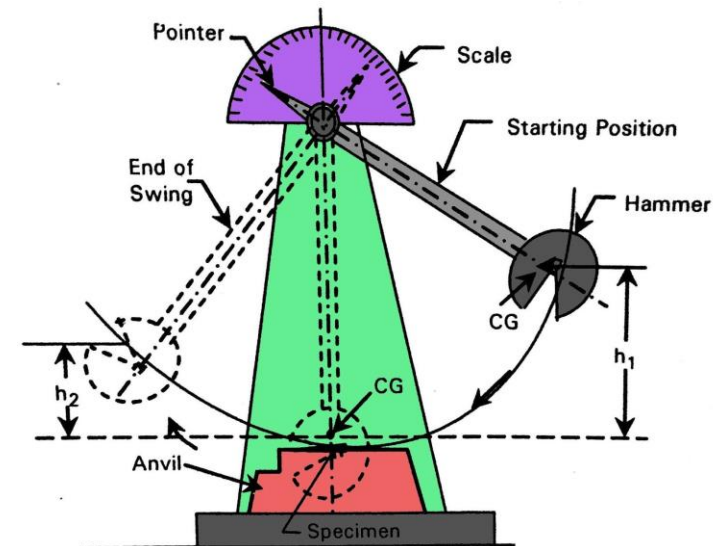
Hardness Test

- Resistance to localized plastic deformation.
- Proportional to tensile strength.
- Cheap, easy, and non-destructive to test.



Impact Test

- It's a test of the ability of a material to withstand impact, used by engineers to predict its behaviour under actual conditions.
- Many materials fail suddenly under impact, at flaws, cracks, or notches.



Variability of Material Properties

Computation of average value:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Computation of standard deviation:

$$s = \left[\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} \right]^{1/2}$$

where n is the number of observations or measurements and x_i is the value of a discrete measurement.

Design/Safety Factors

- For less critical static situations and when tough materials are used.
- A **design stress**, σ_d , is taken as the calculated stress level σ_c (on the basis of the estimated maximum load) multiplied by a *design factor* (N'), that is;

$$\sigma_d = N' \sigma_c$$

- A **safe stress** or *working stress* (σ_w), is used instead of design stress:

$$\sigma_w = \frac{\sigma_y}{N} \quad N = \text{is a factor of safety.}$$

Example

- A tensile-testing apparatus is to be constructed that must withstand a maximum load of 220,000 N. The design calls for two cylindrical support posts, each of which is to support half of the maximum load. Furthermore, plain-carbon (1045) steel ground and polished shafting rounds are to be used; the minimum yield and tensile strengths of this alloy are 310 MPa and 565 MPa, respectively. Specify a suitable diameter for these support posts.

Summary

<i>Equation</i>	<i>Solving For</i>
$\sigma = \frac{F}{A_0}$	Engineering stress
$\epsilon = \frac{l_i - l_0}{l_0} = \frac{\Delta l}{l_0}$	Engineering strain
$\sigma = E\epsilon$	Modulus of elasticity (Hooke's law)
$\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\epsilon_y}{\epsilon_z}$	Poisson's ratio
$\%EL = \left(\frac{l_f - l_0}{l_0} \right) \times 100$	Ductility, percent elongation
$\%RA = \left(\frac{A_0 - A_f}{A_0} \right) \times 100$	Ductility, percent reduction in area
$\sigma_T = \frac{F}{A_i}$	True stress
$\epsilon_T = \ln \frac{l_i}{l_0}$	True strain
$\sigma_T = K\epsilon_T^n$	True stress and true strain (plastic region to point of necking)
$TS(\text{MPa}) = 3.45 \times HB$ $TS(\text{psi}) = 500 \times HB$	Tensile strength from Brinell hardness
$\sigma_w = \frac{\sigma_y}{N}$	Safe (working) stress

Summary Questions

- State all the mechanical properties of metals and define them clearly.

End of the chapter!
Thank you!